

INTRODUCTION TO MACHINE LEARNING

Basic Concepts, Supervised Learning, Unsupervised Learning

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FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Fundamentals of Machine Learning ✓
- 2 Supervised Machine Learning ✓
- 3 Unsupervised Machine Learning ✓

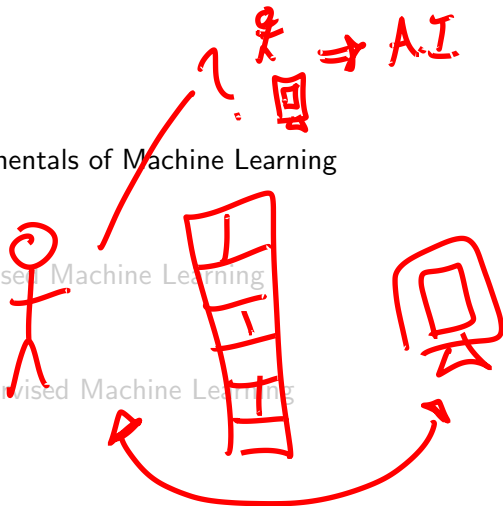


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2 Supervised Machine Learning

3 Unsupervised Machine Learning



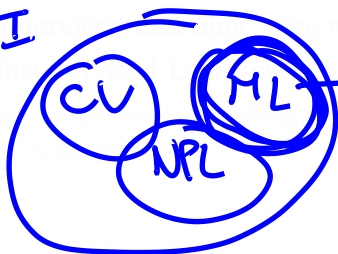
Key Concepts in Machine Learning

 $10^3 \rightarrow 10^6$

Machine Learning

- **Machine learning** is a method of data analysis that **automates** analytical model building.
- It is a **branch** of **artificial intelligence** based on the idea that systems can **learn from data**, **identify patterns** and **make decisions** with minimal human intervention.

AI



data for learning

big volume

\$ 2000 business B.I.

GPT-4.5 $\Rightarrow$ 1 trillion) 250M USD 4 weeks

Key Concepts in Machine Learning

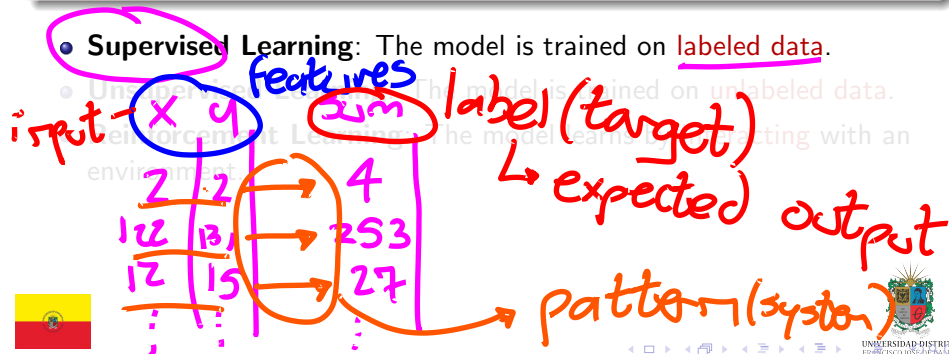
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- **Supervised Learning:** The model is trained on labeled data.

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- **Reinforcement Learning:** The model learns by interacting with an environment.

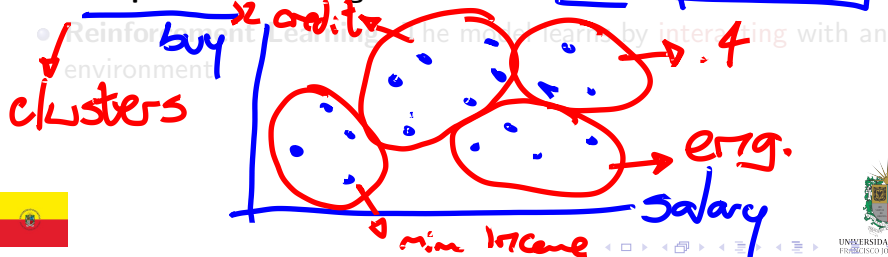


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input → output
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Typical Machine Learning Problems

- **Classification:** Predicting a **label**.
- **Regression:** Predicting a continuous value.
- **Clustering:** Grouping similar data points.
- **Dimensionality Reduction:** Reducing the number of features.
- **Anomaly Detection:** Identifying unusual data points.
- **Association Rule Learning:** Identifying relationships between variables.



→ damage

category

supervised

unsupervised

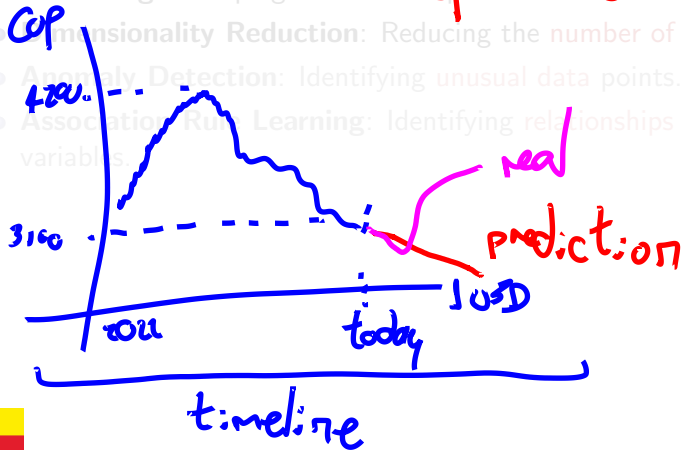


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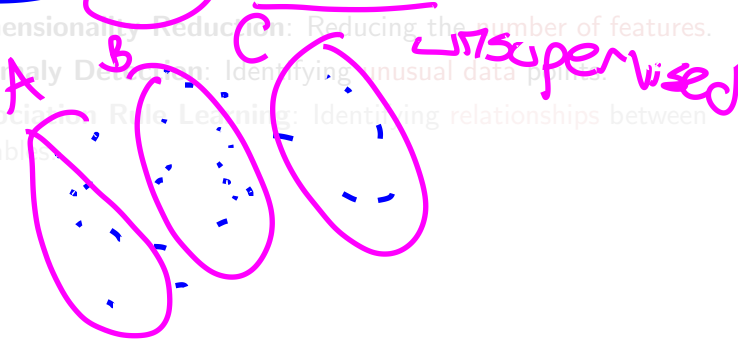
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Handwritten notes above the table: **100D** (red), **1750000** (blue), **200 cols - 300 cols** (blue), **300D** (red).

Handwritten note above the arrow: **Superv. sel** (blue).

Handwritten note next to the reduced table: **50 cols** (pink).

-	-	-	-	-	-



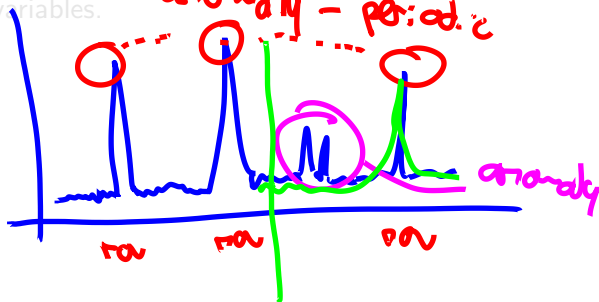
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outlier

unsupervised / regres

anomaly - periodic



Typical Machine Learning Problems

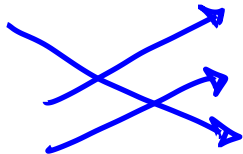
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pre
processing

A



B



morning

↳ sun

↳ wind

rain
afternoon



The Machine Learning Workflow

- **Data Collection:** Gathering the data.

- **Data Preprocessing:** Cleaning and preparing the data.

- **Feature Engineering:** Creating new features.

- **Model Selection:** Choosing the best model.

- **Model Training:** Training the model on the data.

- **Model Evaluation:** Assessing the model's performance.

- **Model Deployment:** Putting the model into production.

data sources + ETL + database

↓
data warehouse

data engineers



The Machine Learning Workflow

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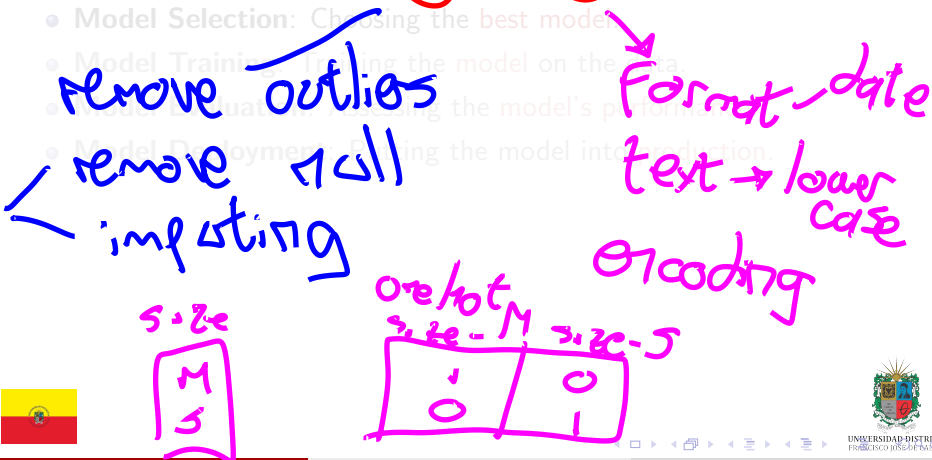
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- **Model Evaluation:** Assessing the model's performance.
- **Model Deployment:** Using the model for prediction.

X

Train	val	test
4.5	2.0	
4.1	4.4	
3.0	3.0	

decr.
↓
0
0



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← **Model Training:** Training the **model** on the data.

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EDA

SUP-

- class

KNN

ANN

- reg

x Boost

;

SUP-

- class

;

- clust.

;



The Machine Learning Workflow

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Handwritten notes illustrating the workflow:

Model Evaluation: Assessing the model's performance.

Model Deployment: Fitting the model into production.

learn (handwritten) → **pattern detection** (handwritten)

Handwritten calculations:

$$\begin{array}{l} 2 \times 2 = 4 \\ 2 \times 3 = 6 \end{array} \quad \begin{array}{l} 3 \times 2 = 6 \\ 3 \times 3 = 9 \end{array} \quad \begin{array}{l} 4 \times 2 = 8 \\ 4 \times 3 = 12 \end{array}$$

$$\begin{array}{l} 2 \times 9 = 18 \\ 3 \times 9 = 27 \end{array} \quad \begin{array}{l} 4 \times 9 = 36 \end{array}$$



The Machine Learning Workflow

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$$2 \times 3 = ? \quad (6)$$

$$4 \times 2 = ? \quad (8)$$

$$4 \times 3 = ? \quad (12)$$

$$3 \times 9 = ? \quad (27)$$

learn $\rightarrow$ less error



The Machine Learning Workflow

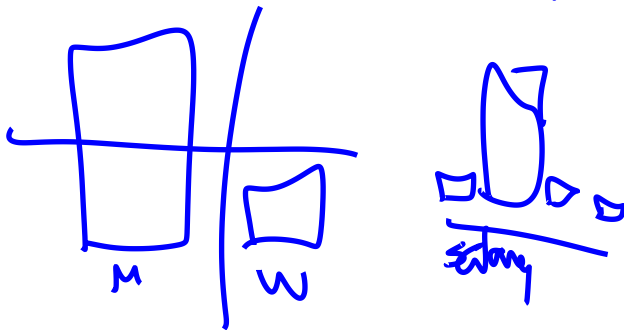
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Algorithmic Bias - sesgo

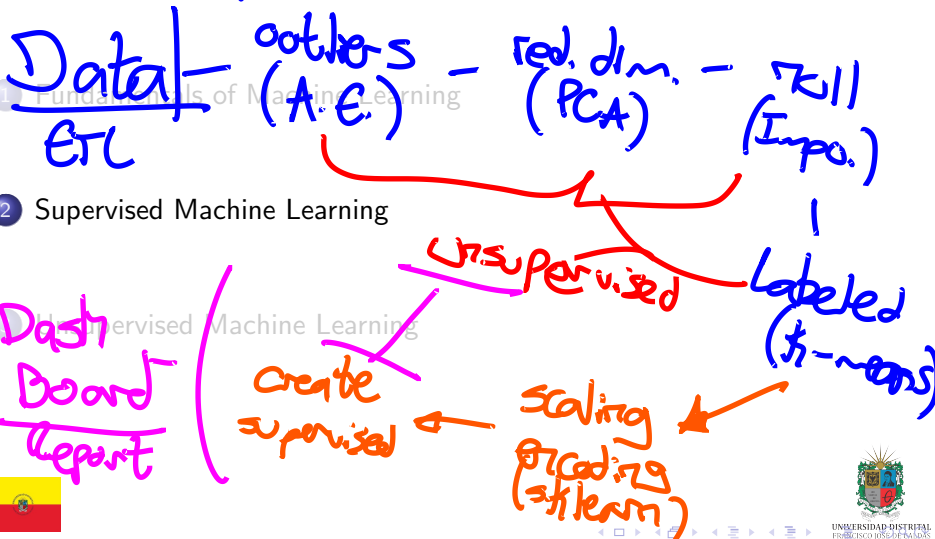
- **Algorithmic bias** is a **systematic error** in a model that results in **unfair outcomes**.
- It can be caused by **biased training data**, biased algorithms, or biased decision-making.

→ avoid → pre-processing



Outline

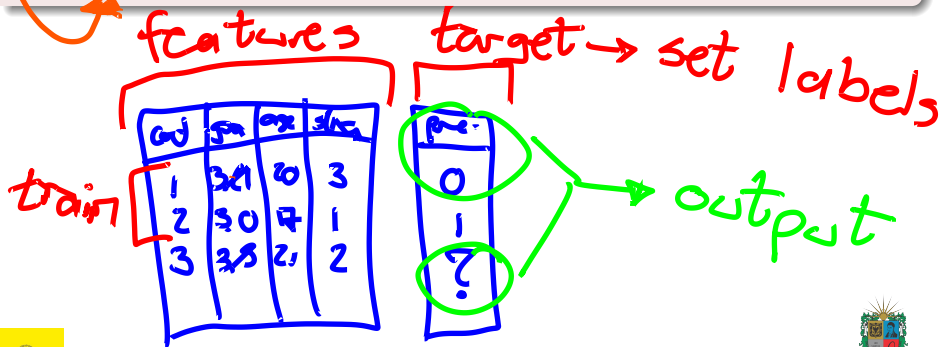
semi-supervised



Introduction to Supervised Machine Learning

Definition

- **Supervised learning** is a type of machine learning where the model is trained on labeled data.
- It involves training a model to map input data to output data based on example input-output pairs.



Overfitting and Underfitting

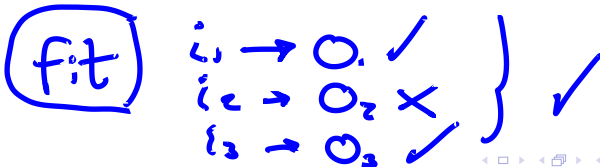
Overfitting

Overfitting occurs when a model learns the training data too well and performs poorly on new data.

Underfitting

Underfitting occurs when a model is too simple to capture the underlying structure of the data.

train ✓ test ✗



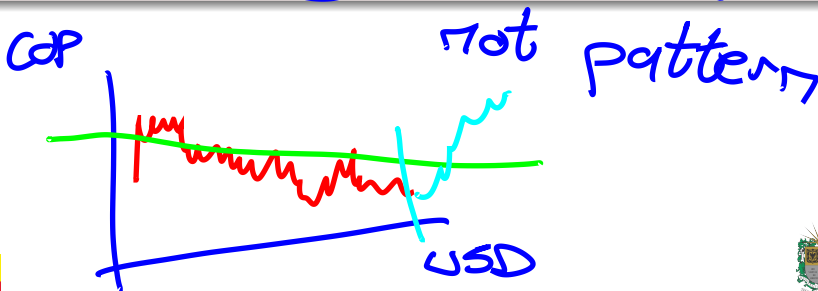
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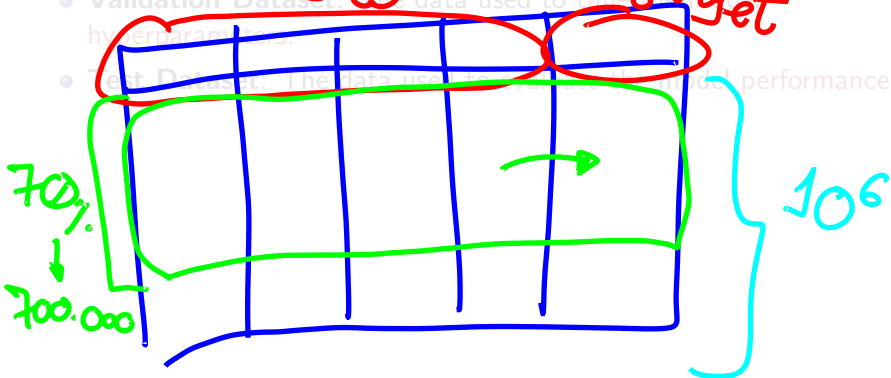
Underfitting occurs when a model is too simple to capture the underlying structure of the data.



Supervised Learning Datasets

- **Training Dataset:** The data used to train the model.

- **Validation Dataset:** The data used to tune the model's hyperparameters.
- **Test Dataset:** The data used to evaluate the model's performance.



learn $\rightarrow$ less errors

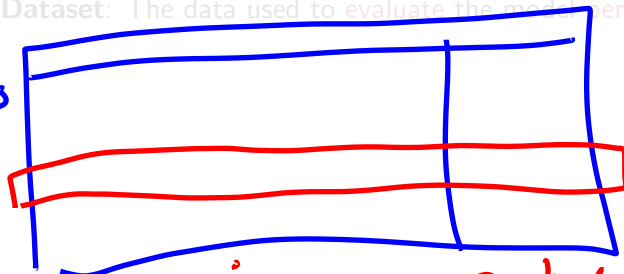


Supervised Learning Datasets

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train
in
libraries

10%
100.000



hyperparameter
tuning

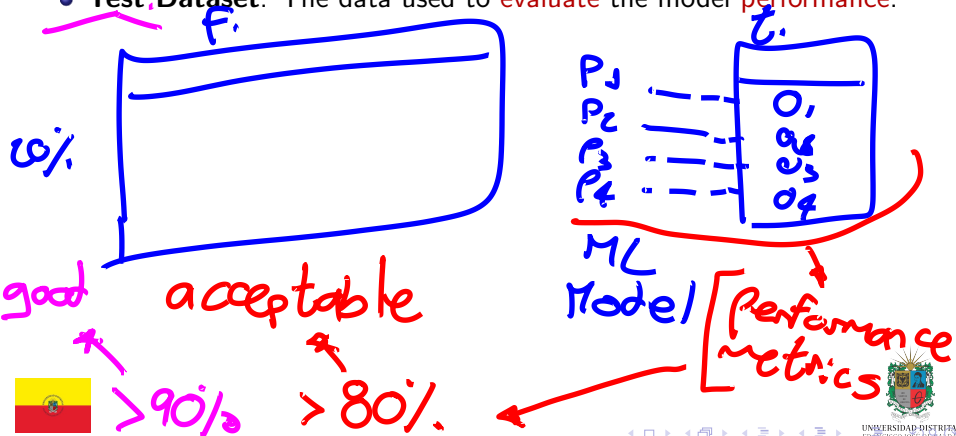
prediction target
error



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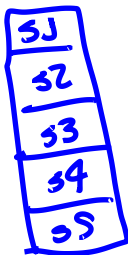
Supervised Learning Datasets

- **Training Dataset:** The data used to **train** the model.
- **Validation Dataset:** The data used to **tune** the model hyperparameters.
- **Test Dataset:** The data used to **evaluate** the model performance.



Cross-Validation

- **Cross-validation** is a technique for **assessing the performance** of a model.
- It involves **splitting** the **data** into multiple **subsets**, training the model on some subsets, and evaluating it on others.
- Common cross-validation **techniques** include **k-fold cross-validation** and **leave-one-out cross-validation**.
- Cross-validation helps to **reduce overfitting** and provides a more **accurate estimate** of the model's **performance**.

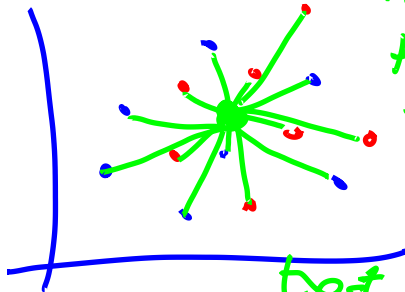


train: s1 s2 s3 s4 test: s5
 train: s2 s3 s4 s5 test: s1
 train: s3 s4 s5 s1 test: s2
 " s4 s5 s1 s2 s3
 " s5 s1 s2 s3



- **K-Nearest Neighbors (KNN)** is a simple algorithm that stores all available cases and classifies new cases based on a similarity measure.
- It can be used for both classification and regression tasks.
- For classification, the output is the class label of the majority of the k-nearest neighbors.

x	y	1
c_1, c_2	r	
c_3, c_4	b	
$\vdots$	$\vdots$	
$\vdots$	$\vdots$	
$\vdots$	$\vdots$	
$\vdots$	$\vdots$	

 $k=1 \rightarrow b$
$$h=3 \rightarrow \Gamma$$
$$k=5 \rightarrow \sqrt{\quad}$$

test - 20. 



fit history

- **Linear regression** is a type of regression analysis used for predicting the value of a continuous dependent variable.
- It works by finding the line that best fits the data.

Least Squares

~~Least squares is a method for finding the best fitting line by minimizing the sum of the squared differences between the predicted and actual values.~~

$$f(x) = \underline{ax} + \underline{b}$$

Linear Regression with Least Squares

Linear Regression

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Least Squares

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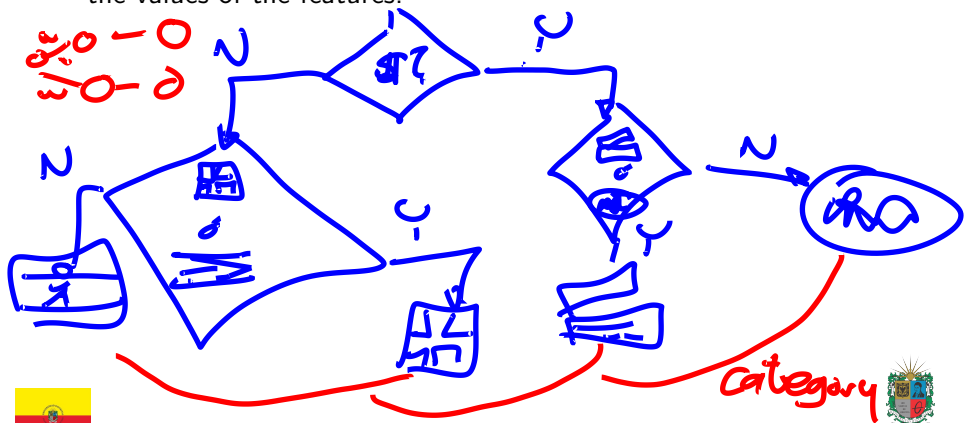


min distance $\sum$
error



Decision Trees

- **Decision trees** are a type of **machine learning model** that can be used for both **classification** and **regression** tasks.
- They work by recursively **partitioning** the **data** into **subsets** based on the values of the features.



Random Forest

100 trees

- **Random forest** is an **ensemble learning** method that combines **multiple decision trees** to create a strong predictive model.
- It works by building **multiple trees** and **averaging** their predictions to **reduce overfitting**.



less error



Neural Networks

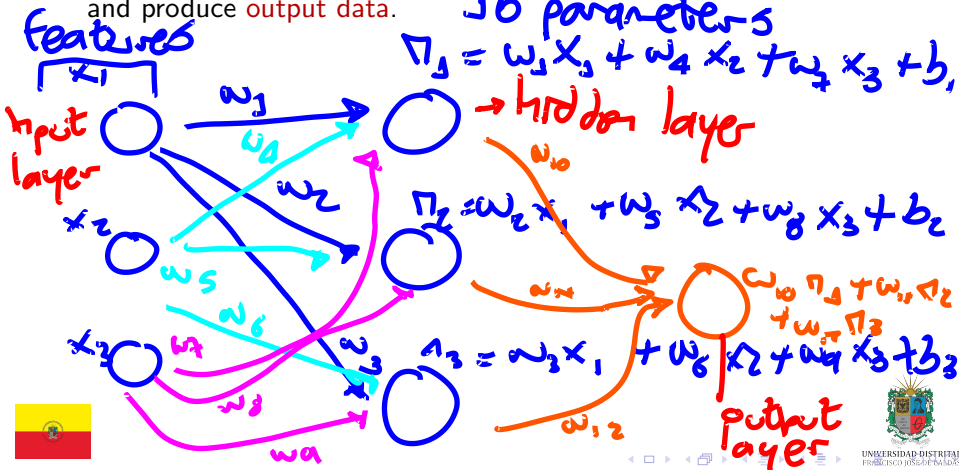
~ 1943 / Perceptron / Backpropagation
 ~ 50 / 70

- **Neural networks** are a type of machine learning model inspired by the human brain.

$$f(x) = ax + b$$

- They consist of layers of interconnected nodes that process input data and produce output data.

16 parameters



Outline

$$\begin{aligned}
 x_0 &= 0 \quad w_1 x_1 + w_2 x_2 + b_1 = x_3 \\
 x_0 &= 0 \quad w_3 x_1 + w_4 x_2 + b_2 = x_4
 \end{aligned}$$

$$\begin{aligned}
 x_0 &= 0 \quad w_5 x_3 + w_6 x_4
 \end{aligned}$$

$$\begin{aligned}
 &w_5 (w_1 x_1 + w_2 x_2) + w_6 (w_3 x_1 + w_4 x_2) \\
 &w_5 w_1 x_1 + w_5 w_2 x_2 + w_6 w_3 x_1 + w_6 w_4 x_2
 \end{aligned}$$

3 Unsupervised Machine Learning

$$\begin{aligned}
 &(w_5 w_1 + w_6 w_3) x_1 + (w_5 w_2 + w_6 w_4) x_2 \\
 &= C_1 x_1 + C_2 x_2
 \end{aligned}$$



Introduction to Unsupervised Machine Learning

Definition

- **Unsupervised learning** is a type of **machine learning** where the model is trained on **unlabeled data**
- It involves finding **patterns** and **relationships** in the data without any predefined labels.

truck
drivers

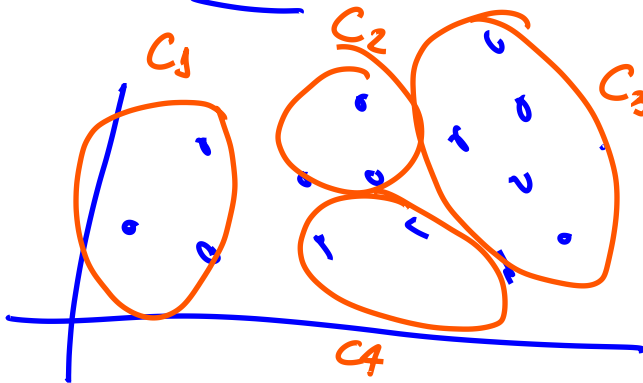
age	city	avg-road	type truck

new label
1
2
3



Clustering

- **Clustering** is a type of **unsupervised learning** that involves **grouping** similar **data points** together.
- Common clustering algorithms include **k-means**, **hierarchical clustering**, and **DBSCAN**.



K-means

Leo Burnett

- **K-means** is a popular **clustering algorithm** that partitions the data into **K distinct clusters** based on feature similarity.
- It works by iteratively assigning data points to the nearest **cluster centroid** and updating the centroids based on the assigned points.



Anomaly Detection

- **Anomaly detection** is a type of **unsupervised learning** that involves identifying **unusual data points** in a dataset.
- Common anomaly detection algorithms include **Isolation Forest**, **One-Class SVM**, and **Autoencoders**.

D.L.

anomalies - outliers



Autoencoders

unbalanced

- **Autoencoders** are a type of **neural network** used for unsupervised learning.
- They work by **encoding** the input data into a lower-dimensional representation and then **decoding** it back to the original data.

ANN → pattern

depth
+
hidden

1 normal

0 anomaly

binary
classification

GANs
Generative
Adversarial
Network



Dimensionality Reduction



- **Dimensionality reduction** is a technique for **reducing** the number of **features** in a dataset while **retaining** as much **information** as possible.
- Common dimensionality reduction techniques include Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE).

mathematical representation

X									
	X								
		X							
			X						
				X					
					X				
						X			
							X		
								X	

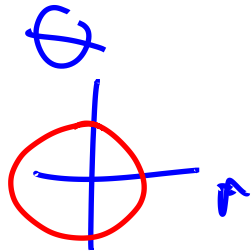
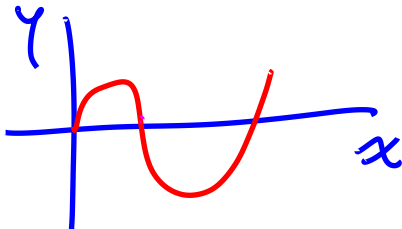
Titanic - 5 cols
 Possum - 20 cols

real-world
 200-300
 rows



Principal Component Analysis (PCA)

- **Principal Component Analysis (PCA)** is a **statistical technique** used for **dimensionality reduction**.
- It works **by transforming** the data into a **new coordinate system** where the greatest variance lies along the first principal **component**, the **second greatest variance** along the second principal **component**, and so on.

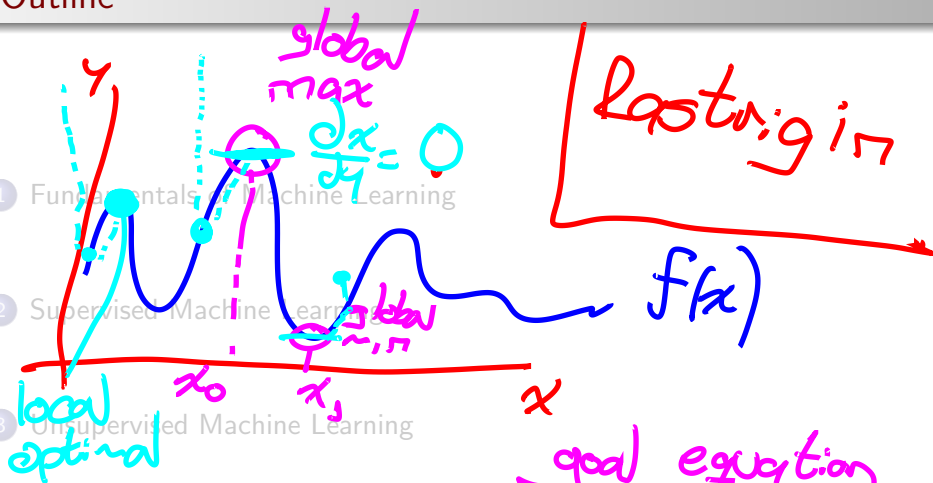


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Thanks!

Backmarks

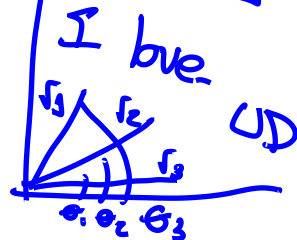
Questions?

I	1
love	2
UD	3



embedding

$1, \theta_1, r_1$
 $2, \theta_2, r_2$
 $3, \theta_3, r_3$



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/machine-learning>

